

Diffusion Models for Risk-Aware Portfolio Optimization

Jihyeong Jeon
Seoul National University
Seoul, South Korea
jeonjihyeong@snu.ac.kr

Jeongyoung Lee
Seoul National University
DeepTrade Technologies Inc.
Seoul, South Korea
ljkleee@snu.ac.kr

U Kang
Seoul National University
Seoul, South Korea
ukang@snu.ac.kr

Abstract

Given a user's risk preference and historical data, how can we generate diverse and high-quality portfolios for risk-aware portfolio optimization? Portfolio optimization is a core financial problem that balances returns and risks by determining asset allocations. While deterministic deep learning approaches can be directly optimized for a single solution, they offer limited flexibility to handle users' risk preferences. In contrast, stochastic methods typically rely on multi-stage processes, which complicate training and do not strictly align with the ultimate goal of generating optimal portfolios.

In this paper, we propose DIFFOLIO (Diffusion Models for Risk-Aware Portfolio Optimization), a novel diffusion-based framework that directly learns a pseudo-optimal portfolio distribution, addressing the drawbacks of (i) deterministic models lacking flexibility and (ii) stochastic models that are complex and misaligned with the true portfolio optimization objective. Instead of forecasting entire future time series, DIFFOLIO directly samples portfolios, immediately adapting to user-specified risk levels through a dedicated risk guidance mechanism embedded in the denoising diffusion process.

Empirical results on multiple real-world market datasets show that DIFFOLIO significantly outperforms existing baselines in terms of return, risk control, and overall reliability. In particular, DIFFOLIO achieves up to 12.1%p higher Annualized Rate of Return, demonstrating its strong potential as a risk-aware and objective-oriented solution to portfolio optimization.

CCS Concepts

• **Applied computing** → *Economics*; • **Information systems** → *Data mining*; • **Mathematics of computing** → *Probabilistic representations*.

Keywords

probabilistic models; deep learning; portfolio optimization; risk management

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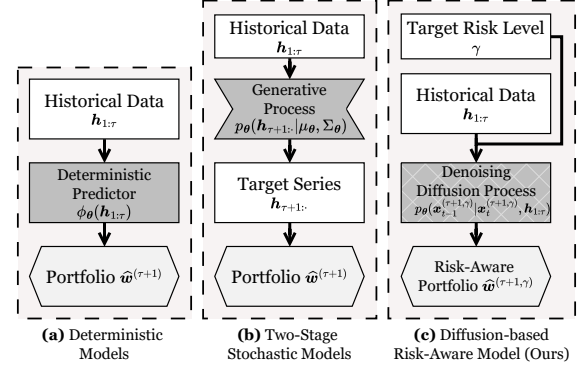


Figure 1: The overview of portfolio optimization methodologies. (a) Deterministic models map historical data $h_{1:T}$ to a single portfolio $\hat{w}^{(T+1)}$. (b) Two-stage stochastic models first generate future time series $h_{T+1:T}$, then construct a portfolio from these forecasts. (c) Our proposed approach, DIFFOLIO, directly samples risk-aware portfolios $\hat{w}^{(T+1,\gamma)}$, conditioning on both the target risk level γ and historical data $h_{1:T}$.

1 Introduction

Given a user's risk preference and historical data, how can we generate diverse, high-quality portfolios for risk-aware portfolio optimization? Portfolio optimization is a cornerstone problem that aims to allocate assets to maximize returns subject to a specified level of risk. Its importance stems from the dual demands of *return maximization*, where investors seek to grow their capital efficiently, and *risk control*, which is critical for mitigating losses in volatile or uncertain market conditions. Because financial markets exhibit complex dynamics and often unpredictable behavior, effectively balancing these two objectives requires models that can adapt to diverse market states as well as varied investor preferences [32].

Existing approaches for portfolio optimization can be broadly classified into *deterministic* and *stochastic* models. Deterministic methods often rely on neural networks that produce a single portfolio aligned with a fixed objective, thereby limiting their ability to handle varying risk tolerances or market uncertainties [46, 57]. Stochastic models address this shortcoming by sampling multiple portfolios, promising more flexible risk management [11, 26]. However, most of these stochastic models adopt a *two-stage* process — first forecasting future prices, then constructing portfolios from these forecasts. Such an indirect procedure often misaligns the learning target with the true optimization objective and increases modeling complexity without yielding a direct gain in performance. Furthermore, although these methods can produce a broader range of outcomes, they lack an explicit mechanism for ensuring that a user-specified target risk is precisely met, leading to insufficient control over the actual risk exposure of the final portfolio.

To address these challenges, we propose DIFFOLIO (**Diff**usion **Models** for **R**isk-Aware **P**ortfolio **O**ptimization), a novel diffusion-based framework for risk-aware portfolio optimization. Instead of forecasting future time series and then constructing portfolios in a two-step fashion, our approach directly learns a *pseudo-optimal portfolio* distribution derived from historical data. This design focuses the model's learning on the actual portfolio optimization objective, allowing immediate sampling of allocations without the need for any intermediate price predictions. In addition, we embed a user-defined target risk level into the sampling procedure by conditioning the reverse diffusion process on the desired risk and introduce a *risk guidance* mechanism to refine denoising trajectories for precise convergence to that risk. As a whole, our framework inherits the direct optimization advantage of deterministic methods together with the versatility of stochastic methods. DIFFOLIO produces multiple solutions and directly targets the portfolio optimization objective, thus bridging the gap between previous deterministic and stochastic methodologies. Figure 1 provides a conceptual overview. Extensive experiments demonstrate that DIFFOLIO consistently produces allocations that accurately reflect user-specified risk levels and surpasses existing baselines in portfolio optimization tasks, achieving the state-of-the-art performance.

Our Main Contributions are summarized as follows:

- We propose DIFFOLIO, a novel diffusion-based deep generative framework for risk-aware portfolio optimization.
- We introduce a risk-conditioning mechanism that offers user-specific control over portfolio risk levels during generation. We also develop a sampling technique within the denoising step to refine risk management in the generated portfolios.
- Through extensive experiments on diverse market datasets, we show that DIFFOLIO outperforms existing methods, achieving up to 12.1%p higher Annualized Rate of Return than the state-of-the-art baselines. These results highlight DIFFOLIO as a promising approach for real-world portfolio optimization.

2 Background

Diffusion Probabilistic Model. Diffusion models [16, 39] have been widely used in deep generative modeling. They are hierarchical latent variable models that define a sequence of latent variables $\mathbf{x}_1, \dots, \mathbf{x}_T$, each having the same dimensionality as the observed data $\mathbf{x}_0 \sim q(\mathbf{x}_0)$. These latent variables gradually encode data through a series of transformations, enabling a generative process that maps Gaussian noise to the data distribution.

The model defines two Markov processes: the forward process $q(\mathbf{x}_{1:T} | \mathbf{x}_0)$ and the reverse process $p_\theta(\mathbf{x}_{0:T})$. The forward process is defined with $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$, where the noise scale β_t is fixed according to a diffusion schedule $\{\beta_t\}_{t=1}^T \in (0, 1)^T$. Conversely, the reverse process is parameterized with $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$, which aims to sequentially reconstruct the original data starting from Gaussian noise. The prior for the reverse process is initialized as $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$. Both processes can be factorized as

$$\begin{aligned} q(\mathbf{x}_{1:T} | \mathbf{x}_0) &:= \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}) \\ p(\mathbf{x}_{0:T}) &:= p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \end{aligned} \quad (1)$$

To optimize the reverse process, we minimize the negative log-likelihood of the data $-\log p_\theta(\mathbf{x}_0)$. Since direct computation of this term is intractable, the variational principle is employed to derive a variational bound:

$$\begin{aligned} -\log p_\theta(\mathbf{x}_0) &= -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \\ &\leq \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right] =: \mathcal{L}_{\text{VLB}}(\mathbf{x}_0). \end{aligned} \quad (2)$$

The variational lower bound $\mathcal{L}_{\text{VLB}}(\mathbf{x}_0)$ can be decomposed as:

$$\mathcal{L}_{\text{VLB}}(\mathbf{x}_0) = \mathcal{L}_0(\mathbf{x}_0) + \mathcal{L}_T(\mathbf{x}_0) + \sum_{t>1} \mathcal{L}_{t-1}(\mathbf{x}_0), \quad (3)$$

where $\mathcal{L}_0(\mathbf{x}_0) = \mathbb{E}_q[-\log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)]$, $\mathcal{L}_T(\mathbf{x}_0)$ is the KL divergence between the prior $p(\mathbf{x}_T)$ and the distribution of $\mathbf{x}_T$ under the forward process, and $\mathcal{L}_{t-1}(\mathbf{x}_0) = \mathbb{E}_q[\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]$. In practice, $\mathcal{L}_T(\mathbf{x}_0)$ is often negligible if $q(\mathbf{x}_T) \approx p(\mathbf{x}_T)$ under an appropriate diffusion schedule [16].

An essential property of the forward process is that samples $\mathbf{x}_t$ can be directly expressed in terms of $\mathbf{x}_0$ as:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (4)$$

where $\alpha_t := (1 - \beta_t)$ and $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$. This allows efficient sampling from intermediate timesteps without explicitly simulating the forward process for all preceding steps. With this, we derive the tractable forward process posterior:

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_q(t), \Sigma_q(t)), \quad (5)$$

where $\boldsymbol{\mu}_q(t) = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$, $\Sigma_q(t) = \sigma_q^2(t) \mathbf{I}$, and $\sigma_q^2(t) = \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}$ are computed from the forward process.

The optimization objective is reformulated to directly minimize the error between the predicted $\hat{\mathbf{x}}_\theta$ and original sample $\mathbf{x}_0$ as:

$$\begin{aligned} \arg \min_{\theta} \sum_{t=1}^T \mathbb{E}_q[\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))] \\ = \arg \min_{\theta} \sum_{t=1}^T \frac{1}{2\sigma_q^2(t)} \frac{\bar{\alpha}_{t-1}(1 - \alpha_t)^2}{(1 - \bar{\alpha}_t)^2} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|_2^2], \end{aligned} \quad (6)$$

where $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$ represents the estimate of the original data.

To simplify the objective, the coefficient terms are omitted, focusing the loss evaluation more on the later diffusion timesteps where uncertainty is higher and precise modeling is more critical.

$$\mathcal{L}(\theta) := \mathbb{E}_{\mathbf{x}_t, \mathbf{x}_0, \epsilon} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|_2^2], \quad (7)$$

where $\mathbf{x}_t$ is directly sampled using the forward process in Eq. (4).

Risk-Aware Portfolio Optimization. We give a formal definition of the risk-aware portfolio optimization problem as follows:

PROBLEM 1 (RISK-AWARE PORTFOLIO OPTIMIZATION).

Given:

- *Historical price data* $\mathcal{X} = \{\mathbf{X}_{\tau,n} | \tau = 1, \dots, T_d; n = 1, \dots, N\}$, where T_d is the total number of trading days and $\mathbf{X}_{\tau,n} \in \mathbb{R}^{L \times F}$ denotes the F -dimensional price features of asset n at trading step τ (market open) over L look-up days.
- *Historical market index* $\mathcal{G} = \{\mathbf{G}_\tau | \tau = 1, \dots, T_d\}$, where each $\mathbf{G}_\tau \in \mathbb{R}^{L \times F}$ denotes the F -dimensional price features of the benchmark index over L look-up days.
- *Target risk level* $\gamma \in \{0, 1, \dots, \gamma_{\max} - 1\}$, an integer specifying the acceptable risk threshold, where $\gamma = 0$ corresponds to the

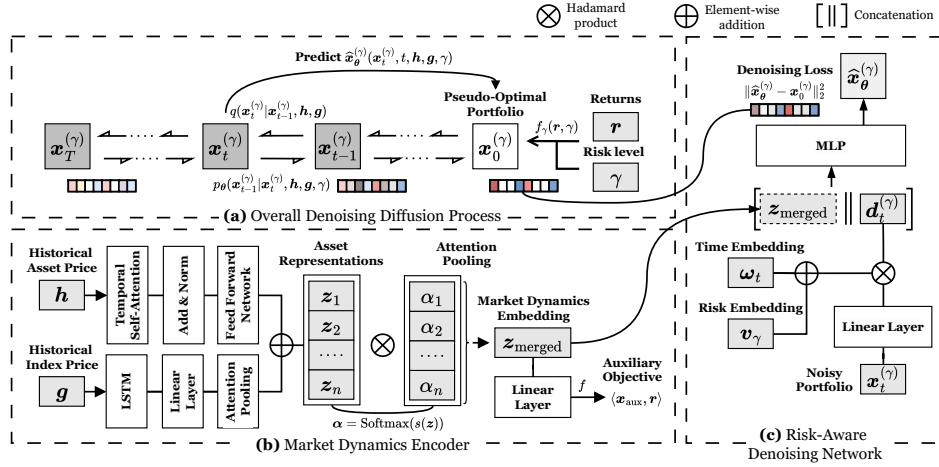


Figure 2: The framework of the proposed DIFFOLIO, a diffusion model for risk-aware portfolio optimization.

safest portfolio and increasing values of γ allow for progressively higher risk levels. In this work, we consider $\gamma_{\max} = 5$.

Find: A sequence of portfolio weights $\mathbf{w}^{(\tau, \gamma)} = [w_1^{(\tau, \gamma)}, \dots, w_N^{(\tau, \gamma)}]^\top \in \mathbb{R}^N$ for each trading step τ , where $w_n^{(\tau, \gamma)}$ represents the proportion of capital allocated to asset n at time τ with target risk level γ .

Objective: Maximize the cumulative portfolio return $\mathcal{R}(\mathbf{w})$ over the trading horizon, defined as:

$$\mathcal{R}(\mathbf{w}^{(\tau, \gamma)}) = \sum_{\tau=1}^{T_d} \mathbf{w}^{(\tau, \gamma)\top} \mathbf{r}_\tau, \quad (8)$$

where $\mathbf{r}_\tau \in \mathbb{R}^N$ is the target returns for trading step τ , subject to the following constraints:

- **Risk constraint:** Ensure that the risk of the portfolio sequence $\mathbf{w}^{(\tau, \gamma)}$, measured by the risk metric $\rho(\cdot)$, lies within the bounds defined by adjacent risk levels [1], i.e.,

$$\rho(\mathbf{w}^{(\tau, \gamma-1)}) < \rho(\mathbf{w}^{(\tau, \gamma)}) < \rho(\mathbf{w}^{(\tau, \gamma+1)}), \quad \forall \gamma, \tau, \quad (9)$$

where $\rho(\mathbf{w}^{(\tau, \gamma)})$ is a risk of the portfolio sequence with target risk level γ . In this work, we define $\rho(\cdot)$ as the Annualized Volatility. For the edge cases, the following boundary conditions apply:

$$\rho(\mathbf{w}^{(\tau, 0)}) < \rho(\mathbf{w}^{(\tau, 1)}) \quad \text{and} \quad \rho(\mathbf{w}^{(\tau, \gamma_{\max}-2)}) < \rho(\mathbf{w}^{(\tau, \gamma_{\max}-1)}). \quad (10)$$

- **Weight constraint:** The portfolio weights must satisfy the following conditions:

$$\sum_{n=1}^N |w_n^{(\tau, \gamma)}| = 1 \quad \text{and} \quad -1 \leq w_n^{(\tau, \gamma)} \leq 1, \quad \forall n, \tau, \gamma \quad (11)$$

where a negative weight, $w_n^{(\tau, \gamma)} < 0$, indicates a short position.

3 Proposed Method

In this section, we present DIFFOLIO, a novel diffusion-based framework for risk-aware portfolio optimization. Our approach addresses the following key challenges:

- C1 Objective-Oriented Generative Modeling.** Many existing methods either (i) rely on deterministic single-portfolio outputs that lack diversity, or (ii) adopt a multi-stage approach misaligned with the true optimization objective, thus hindering performance. *How can we model a generative process that produces near-optimal portfolios from historical data?*
- C2 Discrete Risk Level Control.** While diversification via stochastic sampling helps manage uncertainty, existing approaches

often neglect the explicit incorporation of user-defined risk levels, limiting controllability. *How can we ensure each generated portfolio aligns with a specific, user-defined risk tolerance?*

- C3 Fine-Grained Risk Control.** Even if a model can produce portfolios at different discrete risk levels, deviations during the sampling process may cause the actual risk to drift. *How can we refine the generation so that portfolios remain close to the desired risk throughout the sampling process?*

To address these challenges, DIFFOLIO unifies the following three core ideas into an end-to-end diffusion framework for risk-aware portfolio optimization:

- I1 Diffusion Models for Portfolio Optimization (§3.1).** We define a *pseudo-optimal portfolio distribution* aligned with the optimization objective and train a denoising diffusion process on it, enabling direct generation of candidate portfolios. An *auxiliary objective* further refines the model's conditional representations for market dynamics.
- I2 Risk-Aware Denoising Network (§3.2).** We enable discrete risk level control by training on synthesized, risk-dependent pseudo-optimal portfolios. A dedicated network $\mathcal{F}_\gamma$ maps noisy samples to allocations that match each risk level, ensuring the final outputs align with user-specified risk criteria.
- I3 Denoising Diffusion with Risk Guidance (§3.3).** During inference, we refine the risk through a gradient-based mechanism, called *risk guidance*, that adjusts each reverse step using a risk proxy. This effectively nudges the generated portfolios toward the target risk level.

Figure 2 summarizes how DIFFOLIO integrates these components into a cohesive framework.

3.1 Diffusion Models for Portfolio Optimization

We begin by exploring the methodology of conditional diffusion modeling for the portfolio optimization problem. First, in pursuit of our ultimate objective of portfolio optimization, we synthesize *pseudo-optimal portfolios* using the existing historical data. Next, we formally define the conditional diffusion model, leveraging the *market dynamics representation* extracted by the learned encoder. Subsequently, we provide optimization objective for an end-to-end training with an additional *auxiliary loss* to aid the learning process.

Pseudo-optimal Portfolio Synthesis. We first synthesize a *pseudo-optimal* portfolio using the given historical price data to train a generative model that can sample near-optimal allocations under varying conditions. Specifically, for each of the N assets at trading step τ , we define the return vector $\mathbf{r}_\tau \in \mathbb{R}^N$ by $\mathbf{r}_\tau = \frac{\mathbf{p}_{\tau+1} - \mathbf{p}_\tau}{\mathbf{p}_\tau}$, where $\mathbf{p}_\tau$ denotes the opening prices at step τ . We then define the pseudo-optimal portfolio $\mathbf{x}^{(\tau)}$ via the normalization function $f : \mathbb{R}^N \rightarrow \mathbb{R}^N$:

$$\mathbf{x}^{(\tau)} = f(\mathbf{r}_\tau) = \frac{\mathbf{r}_\tau}{\sum_{n=1}^N |\mathbf{r}_{\tau,n}|}, \quad \forall \tau. \quad (12)$$

This allocation assigns larger absolute weights to assets with higher-magnitude returns, while still granting smaller weights to underperforming assets. The resulting soft-label-like effect [15], which helps to capture uncertainty when these pseudo-optimal portfolios are used as training data.

Conditional Diffusion Model. We consider a datapoint $\mathbf{x}_0 := \mathbf{x}$ from the collection of *pseudo-optimal* portfolios $\{\mathbf{x}^{(\tau)}\}_{\tau=1}^{T_d}$, each associated with corresponding historical asset price data $\mathbf{h} := \{\mathbf{X}_{\tau,n}\}_{n=1}^N \in \mathbb{R}^{N \times L \times F}$ and an index price data $\mathbf{g} := \mathbf{G}_\tau \in \mathbb{R}^{L \times F}$. Following the standard framework of conditional diffusion models [10, 17], we model the conditional data distribution as $q(\mathbf{x}_0 | \mathbf{h}, \mathbf{g})$, the forward process as $q(\mathbf{x}_{1:T} | \mathbf{x}_0, \mathbf{h}, \mathbf{g})$ that progressively adds noise to $\mathbf{x}_0$, and the reverse process as $p_\theta(\mathbf{x}_{0:T} | \mathbf{h}, \mathbf{g})$ that learns to invert this noising procedure. In turn, our setup provides the flexibility to generate diverse allocations under varying market conditions – addressing the inherent stochasticity of financial data more effectively than deterministic methods.

In classic diffusion frameworks [16, 23], the forward process involves adding Gaussian noise ϵ sampled from a standard normal distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ to the data, and the prior distribution $p(\mathbf{x}_T)$ is assumed to follow $\mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$. However, our portfolio weights obey the constraint $\sum_{i=1}^N |w_i| = 1$, and as the number of assets N increases, individual weights become smaller, resulting in a distribution where most values are near zero with a few larger values. Adding noise from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ disrupts the data structure early in the diffusion process, and the mismatch between the prior and constrained data distribution complicates the reverse process [28, 33].

To address this, we attune the noise variance and prior distribution to match the variance of the training data. Specifically, we define the noise distribution as $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma_x^2 \mathbf{I})$, where $\sigma_x = \sqrt{\mathbb{E}_{\tau,n}[(x_n^{(\tau)} - \mu_x)^2]}$ is the standard deviation over all observed portfolios $\mathbf{x}_0$. This adjustment ensures that the injected noise is proportional to the scale of the data, preserving the signal-to-noise ratio and allowing the data structure to persist longer in the forward process [33]. Additionally, we set the prior distribution to

$$p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mu_x \mathbf{1}, \sigma_x^2 \mathbf{I}),$$

aligning it with the scale of the data and reducing the mismatch between the prior and the learned data distribution. This alignment facilitates a more stable and effective reverse process [28], particularly for constrained data like portfolio weights. In practice, our empirical data suggest $\mu_x \approx 0$, so we set $\mu_x = 0$ to keep theoretical and implementation simplicity.

Market Dynamics Encoder. As shown in [48, 54], an effective modeling of conditional information is vital when dealing with

conditional diffusion models. For this purpose, we design *Market Dynamics Encoder* $\mathcal{E} : (\mathbf{h}, \mathbf{g}) \mapsto \mathbf{z}_{\text{merged}}$, which captures intra-asset correlations over time and the global state using attention-based mechanisms to generate a market dynamics embedding.

Specifically, for historical data $\mathbf{h} \in \mathbb{R}^{N \times L \times F}$, we replace the standard linear query-key-value mappings in self-attention [45] with a *temporal convolution network* $\mathcal{T}_\theta$ [4] that applies two dilated 1D convolutions [52] per asset along the time dimension L . This design provides an exponentially large receptive field while remaining computationally efficient, as dilated convolutions capture long-range dependencies and facilitate parallel processing compared to RNN-based methods [7, 56]. We employ shared convolution kernels across assets, defining the query, key, and value as:

$$\mathbf{Q} = \mathcal{T}_{\theta_q}(\mathbf{h}), \quad \mathbf{K} = \mathcal{T}_{\theta_k}(\mathbf{h}), \quad \mathbf{V} = \mathcal{T}_{\theta_v}(\mathbf{h}),$$

where $\theta_q, \theta_k, \theta_v$ are the parameters of the convolution networks, and zero-padding preserves the original shape in $\mathbb{R}^{N \times L \times F}$.

We flatten feature and time dimension of the result tensor into $d = L \times F$, obtaining $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d}$. We then get an affinity matrix:

$$S_{i,j} = \frac{1}{\sqrt{d}} \sum_{k=1}^d Q_{i,k} K_{j,k},$$

for $i, j \in \{1, \dots, N\}$. A softmax is applied row-wise to yield $\tilde{\mathbf{S}} = \frac{\exp(S_{i,j})}{\sum_{j'=1}^N \exp(S_{i,j'})}$, and the attention output is

$$\tilde{\mathbf{h}}_{i,\cdot} = \sum_{j=1}^N \tilde{S}_{i,j} \mathbf{V}_{j,\cdot}, \quad \forall i,$$

where each row embeds correlated asset representations in context of historical prices of the assets. Recovering back to $\mathbb{R}^{N \times L \times F}$, a residual connection and LayerNorm [2] over feature and temporal dimension complete the correlated asset representation

$$\mathbf{h}_{\text{final}} = \text{LayerNorm}(\mathbf{h}' + \mathcal{T}_{\theta_h}(\mathbf{h}')),$$

where $\mathbf{h}' = \text{LayerNorm}(\mathbf{h} + \tilde{\mathbf{h}})$ and $\mathcal{T}_{\theta_h}$ is a temporal convolution network consisting of two dilated 1D convolutions that preserve the input shape via padding.

In parallel, we extract the global state from $\mathbf{g} \in \mathbb{R}^{L \times F}$ using a recurrent-based attention pipeline [3]. Let

$$\{\mathbf{u}_l\}_{l=1}^L = \text{LSTM}(\mathbf{g}), \quad \mathbf{u}_l \in \mathbb{R}^{d_{\text{hid}}}$$

be the hidden states from the LSTM [18]. Then, an attention mechanism uses a learnable scoring function $\psi : \mathbb{R}^{d_{\text{hid}}} \rightarrow \mathbb{R}$ to score $\mathbf{u}_l$, normalizes via softmax over $l = 1, \dots, L$ to obtain weights α_l , and computes the global representation as their weighted sum:

$$\mathbf{g}' = \sum_{l=1}^L \alpha_l \mathbf{u}_l, \quad \text{where } \alpha_l = \frac{\exp(\psi(\mathbf{u}_l))}{\sum_{l'=1}^L \exp(\psi(\mathbf{u}_{l'}))}. \quad (13)$$

Finally, we linearly project $\mathbf{g}' \in \mathbb{R}^{d_{\text{hid}}}$ to $\mathbf{g}_{\text{final}} \in \mathbb{R}^d$, capturing a global market view. Then it is combined with the per-asset representation $\mathbf{h}_{\text{final}} \in \mathbb{R}^{N \times d}$ to integrate local and global market semantics. For each asset n , the final asset representation $\mathbf{z}_n$ is computed as:

$$\mathbf{z}_n = \mathbf{W}_h \mathbf{h}_{\text{final}}^n + \mathbf{g}_{\text{final}},$$

where $\mathbf{h}_{\text{final}}^n \in \mathbb{R}^d$ denotes the final representation of asset n , and $\mathbf{W}_h \in \mathbb{R}^{d \times d}$ represents a learnable parameter matrix. We then apply an attention-based pooling step, as in Eq. (13), which employs a learned scoring function $s : \mathbb{R}^d \rightarrow \mathbb{R}$ to compute weights $\alpha =$

Softmax($\{s(z_1), \dots, s(z_N)\}$), then merges them to produce market dynamics embedding:

$$z_{\text{merged}} = \sum_{n=1}^N \alpha_n z_n \in \mathbb{R}^d.$$

In this work, we use a simple MLP for the scoring function s .

Optimizing with Auxiliary Loss. As shown in Eq. 7, the optimization objective of the diffusion model reduces to accurately predicting the undiffused portfolio $\hat{\mathbf{x}}_\theta \approx \mathbf{x}_0 \in \mathbb{R}^N$ from a noisy portfolio $\mathbf{x}_t \in \mathbb{R}^N$. To achieve this, we first apply a linear mapping from the noisy portfolio to a d -dimensional space for expressiveness, then infuse timestep information for time awareness as

$$\mathbf{d}_t = \mathbf{W} \mathbf{x}_t \odot \boldsymbol{\omega}_t \in \mathbb{R}^d,$$

where $\odot$ denotes the Hadamard product, $\mathbf{W} \in \mathbb{R}^{d \times N}$ represents learnable weights, and $\boldsymbol{\omega}_t = \sigma(\mathbf{W}_\omega \phi_{\text{pos}}(t))$ is a time embedding for diffusion timestep t , obtained by applying a nonlinear transformation to the sinusoidal embedding $\phi_{\text{pos}}(t) \in \mathbb{R}^d$ [14] with learnable parameters $\mathbf{W}_\omega \in \mathbb{R}^{d \times d}$.

Subsequently, we fuse asset representations $z_{\text{merged}} \in \mathbb{R}^d$ from the market encoder $\mathcal{E}(\mathbf{h}, \mathbf{g})$ with $\mathbf{d}_t \in \mathbb{R}^d$ and define:

$$\mathbf{u}_{\text{cat}} = [z_{\text{merged}} \parallel \mathbf{d}_t] \in \mathbb{R}^{2d},$$

where $[\cdot \parallel \cdot]$ denotes vector concatenation. This is passed through a multi-layer perceptron, $\text{MLP} : \mathbb{R}^{2d} \rightarrow \mathbb{R}^d$ to form a portfolio:

$$\hat{\mathbf{x}}_\theta = \text{MLP}(\mathbf{u}_{\text{cat}}) \in \mathbb{R}^N.$$

We refer to this module, which yields the predicted portfolio $\hat{\mathbf{x}}_\theta$ at $\mathbf{x}_t$, as a *denoising network*, denoted as $\mathcal{F} : (\mathbf{x}_t, t, \mathbf{h}, \mathbf{g}) \mapsto \hat{\mathbf{x}}_\theta$.

For optimization, we iteratively sample noisy portfolio $\mathbf{x}_t$ and minimize the mean-squared error loss in Eq. (7). Furthermore, to facilitate representation learning in the market dynamics encoder $\mathcal{E}$, we maximize an auxiliary objective defined as the portfolio return:

$$\mathcal{L}_{\text{aux}} = \langle f(\mathbf{W}_p \mathcal{E}(\mathbf{h}, \mathbf{g})), \mathbf{r} \rangle, \quad (14)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner-product, f is the normalizer from Eq. (12), $\mathbf{W}_p \in \mathbb{R}^{N \times d}$ maps to portfolio space, and $\mathbf{r} \in \mathbb{R}^N$ is returns associated with $\mathbf{h}$. In summary, we minimize the following objective:

$$\mathcal{L}'(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \epsilon \sim \mathcal{N}(0, \sigma_{x^2} \mathbf{I})} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t, \mathbf{h}, \mathbf{g}) - \mathbf{x}_0\|_2^2] - \lambda \mathcal{L}_{\text{aux}},$$

where λ is a hyperparameter controlling the auxiliary loss weight.

3.2 Risk-Aware Denoising Network

We introduce a *risk-aware denoising network* $\mathcal{F}_\gamma$ that learns distinct denoising trajectories tailored to a user's target risk level γ , thereby promoting effective risk management. To achieve this, we redefine the pseudo-optimal portfolio based on risk levels and incorporate risk embeddings into the denoising network to learn its distribution.

Specifically, to model conditional denoising paths for each target risk level $\gamma \in \{0, 1, \dots, \gamma_{\text{max}} - 1\}$, we first synthesize a risk-dependent pseudo-optimal portfolio $\mathbf{x}^{(\tau, \gamma)}$, designed to reflect sparse, leveraged allocations at higher risk levels and more diversified allocations at lower ones. Namely, we define

$$\mathbf{x}^{(\tau, \gamma)} = f_\gamma(\mathbf{r}_\tau, \gamma), \quad \forall \tau, \gamma, \quad (15)$$

where $\mathbf{r}_\tau \in \mathbb{R}^N$ denotes the returns at trading time τ and f_γ is a risk-dependent normalization function analogous to Eq. (12), selecting only the top- k_γ returning assets. To determine the number k_γ of

selected assets per risk level, we compute:

$$k_\gamma = \left\lfloor \frac{N}{\gamma_{\text{max}}} \right\rfloor (\gamma_{\text{max}} - \gamma),$$

and then define the asset selection set:

$$S^{(\tau, \gamma)} = \text{TopK}(\{ |r_{\tau, n}| \}_{n=1}^N, k_\gamma), \quad (16)$$

i.e., $S^{(\tau, \gamma)}$ is the indices of assets with the k_γ largest absolute returns. We then construct the risk-dependent pseudo-optimal portfolio:

$$x_n^{(\tau, \gamma)} = \begin{cases} \frac{r_{\tau, n}}{Z^{(\tau, \gamma)}}, & \text{if } n \in S^{(\tau, \gamma)}, \\ 0, & \text{otherwise} \end{cases},$$

where $Z^{(\tau, \gamma)} = \sum_{n \in S^{(\tau, \gamma)}} |r_{\tau, n}|$ denotes the normalization constant that ensures the selected assets sum to one in terms of absolute weights. Consequently, only top- k_γ assets receive nonzero weights, ensuring $\sum_{n=1}^N |x_n^{(\tau, \gamma)}| = 1$. For notational brevity, in the following sections we assume an arbitrary trading step and omit τ , denoting the γ risk pseudo-optimal portfolio as $\mathbf{x}_0^{(\gamma)} \in \{\mathbf{x}^{(\tau, \gamma)}\}_{\tau=1}^{|X|}$.

With these risk-dependent pseudo-optimal portfolios, we train risk-aware denoising network $\mathcal{F}_\gamma$ to predict the initial portfolio $\hat{\mathbf{x}}_\theta \approx \mathbf{x}_0^{(\gamma)}$ while incorporating the target risk level γ . In particular, we define the time-risk modulated portfolio embedding as:

$$\mathbf{d}_t^{(\gamma)} = \mathbf{W} \mathbf{x}_t^{(\gamma)} \odot (\boldsymbol{\omega}_t + \mathbf{v}_\gamma), \quad (17)$$

where $\mathbf{v}_\gamma \in \mathbb{R}^d$ is a learnable embedding vector for risk level γ . By concatenating this with asset representation z_{merged} , followed by the MLP (as described in Section 3.1), we obtain the risk-conditioned portfolio prediction $\hat{\mathbf{x}}_\theta^{(\gamma)}$. Here the superscript γ indicates conditioning on the risk level, as with $\mathbf{x}_t^{(\gamma)}$ or $\mathbf{w}^{(\gamma)}$.

Finally, we train $\mathcal{F}_\gamma$ by optimizing both the auxiliary loss $\mathcal{L}_{\text{aux}}$ and the denoising loss between $\hat{\mathbf{x}}_\theta^{(\gamma)}$ and the clean portfolio $\mathbf{x}_0^{(\gamma)}$:

$$\mathcal{L}''(\theta) = \mathbb{E}_{t, \mathbf{x}_0^{(\gamma)}, \epsilon, \gamma} [\|\hat{\mathbf{x}}_\theta^{(\gamma)}(\mathbf{x}_t^{(\gamma)}, t, \mathbf{h}, \mathbf{g}, \gamma) - \mathbf{x}_0^{(\gamma)}\|_2^2] - \lambda \mathcal{L}_{\text{aux}}, \quad (18)$$

which enables the network to learn robust risk-conditioned denoising paths across varying market conditions and user risk levels.

3.3 Denoising Diffusion with Risk Guidance

We introduce a risk guidance mechanism that steers the denoising path toward a desired risk level, thereby enabling fine-grained control over user-defined risk preferences and mitigating deviations in the sampling process. Specifically, we modify each reverse step to favor allocations with lower *portfolio variance*, serving as a practical *proxy risk* for future volatility.

Proxy Risk with Portfolio Variance. Let $\mathbf{w}^{(\tau, \gamma)} = f(\hat{\mathbf{x}}^{(\tau, \gamma)})$ be the portfolio weights sampled at trading step τ under risk level γ , and let ρ denote the *annualized volatility*, i.e., the standard deviation of realized returns over the T_t inference time trading steps:

$$\rho(\mathbf{w}^{(\cdot, \gamma)}) := \sqrt{\text{Var} \left(\sum_{n=1}^N \mathbf{w}_{\cdot, n}^{(\cdot, \gamma)} \odot \mathbf{r}_{\cdot, n} \right) \times D_Y},$$

where $\mathbf{w}^{(\cdot, \gamma)} \in \mathbb{R}^{T_t \times N}$ and $\mathbf{r} \in \mathbb{R}^{T_t \times N}$ denote the sampled portfolio weights and the returns of N assets over T_t trading steps, respectively. Here, the subscript $\cdot, n$ indicates the n -th column vector. Since future returns are unknown at inference time, directly minimizing ρ is impractical. However, approximating risk is crucial for guiding

the generative process toward effective risk control. To this end, we define a *proxy risk* measure based on *portfolio variance*:

$$\rho'(\mathbf{w}^{(\tau, \gamma)}) = \mathbf{w}^{(\tau, \gamma)\top} \widehat{\Sigma} \mathbf{w}^{(\tau, \gamma)}, \quad (19)$$

where $\widehat{\Sigma}$ is the covariance matrix estimated from historical returns. This measure provides insight into the riskiness of the constructed portfolio $\mathbf{w}^{(\tau, \gamma)}$ at each trading step τ . To simplify notation, we omit τ in what follows and write $\mathbf{w}^{(\tau, \gamma)}$ as $\mathbf{w}^{(\gamma)}$.

Risk Guided Sampling. We leverage the proxy risk measure ρ' to adjust the mean of each reverse transition during inference, guiding the denoising process toward allocations with lower estimated variance. This approach encourages *hedged* portfolios, mitigating exposure to highly correlated or similarly oriented positions and thereby reducing realized return fluctuations.

Formally, from forward process posterior in Eq. 5, the risk level γ , and the variance $\sigma_{\mathbf{x}}^2$, we have the γ risk reverse transition

$$p_{\theta}(\mathbf{x}_{t-1}^{(\gamma)} | \mathbf{x}_t^{(\gamma)}, \mathbf{x}_0^{(\gamma)}, \mathbf{h}, \mathbf{g}, \gamma) = \mathcal{N}(\mathbf{x}_{t-1}^{(\gamma)}; \mu_{\theta}^{(\gamma)}(t), \Sigma_{\theta}(t)),$$

where

$$\mu_{\theta}^{(\gamma)}(t) = \frac{\sqrt{\alpha_t} \beta_t}{1 - \alpha_t} \widehat{\mathbf{x}}_{\theta}^{(\gamma)} + \frac{\sqrt{\alpha_t} (1 - \alpha_{t-1})}{1 - \alpha_t} \mathbf{x}_t^{(\gamma)} \text{ and } \Sigma_{\theta}(t) = \frac{1 - \alpha_{t-1}}{1 - \alpha_t} \sigma_{\mathbf{x}}^2 \beta_t \mathbf{I}.$$

To shift the reverse process toward lower variance, we modify the mean $\mu_{\theta}^{(\gamma)}(t)$ by introducing a gradient-based correction term based on the proxy risk ρ' . Let $\rho'(\mathbf{w}_t^{(\gamma)}) = \mathbf{w}_t^{(\gamma)\top} \widehat{\Sigma} \mathbf{w}_t^{(\gamma)}$ denote the proxy risk at reverse step t , where $\mathbf{w}_t^{(\gamma)} = f(\mathbf{x}_t^{(\gamma)})$ is the portfolio constructed using the normalization function from Eq. (12). We then compute the gradient of the proxy risk, $\nabla_{\mathbf{x}_t^{(\gamma)}} \rho'(\mathbf{w}_t^{(\gamma)})$, which indicates how local changes in $\mathbf{x}_t^{(\gamma)}$ would affect the estimated variance. To guide the mean toward lower risk allocations, we subtract this gradient from $\mu_{\theta}^{(\gamma)}(t)$, scaled by a factor $\zeta^{(\gamma)} \in [0, 1]$:

$$\mu_{\theta}^{(\gamma)'}(t) = \frac{\sqrt{\alpha_t} \beta_t}{1 - \alpha_t} \widehat{\mathbf{x}}_{\theta}^{(\gamma)} + \frac{\sqrt{\alpha_t} (1 - \alpha_{t-1})}{1 - \alpha_t} \mathbf{x}_t^{(\gamma)} - \zeta^{(\gamma)} \nabla_{\mathbf{x}_t^{(\gamma)}} \rho'(\mathbf{w}_t^{(\gamma)}), \quad (20)$$

where $\zeta^{(\gamma)} = 1 - \frac{\gamma}{\gamma_{\max} - 1}$, $\gamma \in \{0, 1, \dots, \gamma_{\max} - 1\}$.

Thus, at lowest risk, $\zeta^{(0)} = 1$ yields a full gradient update favoring lower variance, and at highest risk, $\zeta^{(\gamma_{\max} - 1)} = 0$ disables the gradient correction. Then, our *risk-guided* reverse transition becomes

$$p_{\theta}(\mathbf{x}_{t-1}^{(\gamma)} | \mathbf{x}_t^{(\gamma)}, \mathbf{x}_0^{(\gamma)}, \mathbf{h}, \mathbf{g}, \gamma) = \mathcal{N}(\mathbf{x}_{t-1}^{(\gamma)}; \mu_{\theta}^{(\gamma)'}(t), \Sigma_{\theta}(t)). \quad (21)$$

By repeatedly sampling $\mathbf{x}_{t-1}^{(\gamma)}$ from $p_{\theta}^{(\gamma)}$ — which is *re-centered* to favor lower variance solutions — we effectively guide the final portfolio allocation to align with the target risk level.

3.4 Training and Sampling Portfolio Weights

Algorithm 1 presents the training procedure of DIFFOLIO. Pseudo-optimal portfolios, synthesized from historical data and conditioned on risk level γ , serve as denoising targets. At each iteration, a risk level is uniformly sampled, noise is added to the target portfolio $\mathbf{x}_0^{(\gamma)}$, and the model is trained by minimizing a combination of mean-squared error and an auxiliary objective to promote accurate denoising and robust market-aware representations.

Algorithm 2 illustrates the sampling process at inference. Given a trained denoising network $\mathcal{F}_{\gamma}$, we start from a noise sample $\mathbf{x}_T^{(\gamma)} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathbf{x}}^2 \mathbf{I})$ and iteratively apply reverse transitions. Each

Algorithm 1 Training DIFFOLIO

Input: historical price data $\mathcal{X}$, historical market index $\mathcal{G}$, and risk-dependent pseudo-optimal portfolios $\{\mathbf{x}^{(\tau, \gamma)}\}_{\tau=1}^{|\mathcal{X}|}, \forall \gamma \in \{0, 1, \dots, \gamma_{\max} - 1\}$

Output: a learned denoising network $\mathcal{F}_{\gamma}$

```

1: repeat
2:    $\gamma \sim \text{Uniform}(\{0, 1, \dots, \gamma_{\max} - 1\})$ 
3:    $\mathbf{x}_0^{(\gamma)} \sim \text{Uniform}(\{\mathbf{x}^{(\tau, \gamma)}\}_{\tau=1}^{|\mathcal{X}|})$ 
4:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
5:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
6:    $\mathbf{x}_t^{(\gamma)} \leftarrow \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{(1 - \alpha_t)} \sigma_{\mathbf{x}} \epsilon$ 
7:   Take gradient descent step on
        $\nabla_{\theta} \|\widehat{\mathbf{x}}_{\theta}^{(\gamma)}(\mathbf{x}_t^{(\gamma)}, t, \mathbf{h}, \mathbf{g}, \gamma) - \mathbf{x}_0^{(\gamma)}\|_2^2 - \lambda \nabla_{\theta} \mathcal{L}_{\text{aux}}$ 
8: until convergence

```

Algorithm 2 Sampling portfolio weights with risk guidance

Input: historical asset prices $\mathbf{h}$, historical index prices $\mathbf{g}$, a target risk level $\gamma \in \{0, 1, \dots, \gamma_{\max} - 1\}$, and the learned denoising network $\mathcal{F}_{\gamma}$

Output: risk-aware portfolio weights $\widehat{\mathbf{w}}^{(\gamma)} = f_{\gamma}(\mathbf{x}_0^{(\gamma)}, \gamma)$

```

1:  $\mathbf{x}_T^{(\gamma)} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathbf{x}}^2 \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma_{\mathbf{x}}^2 \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} \leftarrow \mathbf{0}$ 
4:    $\mathbf{w}_t^{(\gamma)} \leftarrow f(\mathbf{x}_t^{(\gamma)})$ 
5:    $\rho'(\mathbf{w}_t^{(\gamma)}) \leftarrow \mathbf{w}_t^{(\gamma)\top} \widehat{\Sigma} \mathbf{w}_t^{(\gamma)}$ 
6:    $\widehat{\mathbf{x}}_{\theta}^{(\gamma)} \leftarrow \mathcal{F}_{\gamma}(\mathbf{x}_t^{(\gamma)}, t, \mathbf{h}, \mathbf{g})$ 
7:    $\zeta^{(\gamma)} \leftarrow 1 - \frac{\gamma}{\gamma_{\max} - 1}$ 
8:    $\mu_{\theta}^{(\gamma)'}(t) \leftarrow \frac{\sqrt{\alpha_t} \beta_t}{1 - \alpha_t} \widehat{\mathbf{x}}_{\theta}^{(\gamma)} + \frac{\sqrt{\alpha_t} (1 - \alpha_{t-1})}{1 - \alpha_t} \mathbf{x}_t^{(\gamma)} - \zeta^{(\gamma)} \nabla_{\mathbf{x}_t^{(\gamma)}} \rho'(\mathbf{w}_t^{(\gamma)})$ 
9:    $\mathbf{x}_{t-1}^{(\gamma)} \leftarrow \mu_{\theta}^{(\gamma)'}(t) + \sqrt{\frac{1 - \alpha_{t-1}}{1 - \alpha_t}} \beta_t \sigma_{\mathbf{x}} \mathbf{z}$ 
10: end for
11: return  $f_{\gamma}(\mathbf{x}_0^{(\gamma)}, \gamma)$ 

```

step integrates a risk guidance mechanism (Section 3.3), which adjusts the transition mean based on the current portfolio's variance, steering the sample toward the target risk level γ for fine-grained control over allocation. Finally, we normalize $\mathbf{x}_0^{(\gamma)}$ via Eq. (15) to obtain the risk-aware portfolio weights $\widehat{\mathbf{w}}^{(\gamma)}$.

4 Experiments

We address the following research questions through experiments:

- RQ1 **Quantitative Assessment (§4.2).** How does DIFFOLIO perform compared to existing portfolio optimization models? Specifically, does it achieve superior performance while maintaining risk awareness?
- RQ2 **Ablation Studies (§4.3).** How robust is DIFFOLIO across different target risk levels? Concretely, does it sustain strong performance when the target level reflects market conditions?
- RQ3 **Effectiveness of Risk Guidance (§4.4).** Are the risk-aware denoising network and risk guidance mechanism effective in modulating risk during the reverse process?
- RQ4 **Visualization of Denoising Paths (§4.5).** Do sampling trajectories exhibit distinctive risk measurements corresponding to the target risk levels as the denoising process progresses?
- RQ5 **Risk-Driven Weight Allocation (§4.6).** Does DIFFOLIO allocate portfolio weights adaptively based on risk levels? Specifically, do higher risk levels correspond to increased allocations in riskier assets and lower risk levels to safer assets?

Table 1: Performance comparison of DIFFOLIO (DF) and baselines using ARR and ASR. Best results are in bold. Classical methods are marked with — to indicate zero variance. Symbols † and ‡ indicate statistically significant improvements (p -value < 0.05, two-tailed t -test) over the best two-step generative and neural network-based deterministic baselines, respectively.

Dataset	U.S. stock market		KR stock market		CN stock market		JP stock market		U.K. stock market		Crypto market	
Methodology	ARR(%)†	ASR†	ARR(%)†	ASR†	ARR(%)†	ASR†	ARR(%)†	ASR†	ARR(%)†	ASR†	ARR(%)†	ASR†
<i>Classical baseline</i>												
BuyHold	6.016± —	0.353± —	−1.450± —	−0.063± —	5.830± —	0.342± —	8.595± —	0.578± —	4.732± —	0.409± —	28.39± —	0.881± —
ARIMA [5]	−1.600± —	−0.160± —	−2.891± —	−0.468± —	−35.22± —	−1.543± —	6.451± —	0.639± —	−5.785± —	−0.260± —	−11.68± —	−0.351± —
BWSL [19]	1.758± —	0.113± —	8.530± —	0.969± —	−9.078± —	−0.547± —	13.53± —	1.191± —	−9.578± —	−0.463± —	−2.398± —	−0.057± —
CSM [36]	3.613± —	0.401± —	−9.773± —	−0.432± —	6.600± —	0.628± —	5.107± —	0.974± —	−1.549± —	−0.094± —	−14.11± —	−0.406± —
<i>Two-step generative</i>												
FactorVAE [11]	−3.745±1.965	−0.297±0.154	3.798±4.413	0.321±0.377	−2.538±2.978	−0.136±0.107	−1.405±4.382	−0.122±0.421	4.754±7.767	0.303±0.538	2.170±6.210	0.815±0.225
D-Va [26]	−0.888±0.837	−0.160±0.143	0.398±0.317	0.070±0.054	12.47±8.657	1.681 ±1.125	2.592±4.222	0.448±0.999	0.828±4.236	0.096±0.508	15.32±4.314	0.816±0.294
TMDM [29]	0.670±0.631	0.101±0.096	2.643±1.607	0.383±0.241	4.511±6.806	0.359±0.564	3.122±4.505	0.548±0.746	0.462±6.458	0.037±0.507	7.237±10.52	0.318±0.474
<i>Deterministic</i>												
MetaTrader [34]	11.57±9.302	0.307±0.277	8.257±6.158	0.489±0.311	19.23±8.923	0.562±0.227	17.24±9.958	0.580±0.338	9.668±3.557	0.463±0.127	32.63±17.23	0.764±0.235
EarnMore [55]	12.82±6.198	0.425±0.206	18.61±4.661	0.807±0.291	22.68±6.160	0.782±0.204	15.05±5.737	0.525±0.332	11.16±4.925	0.718±0.389	39.22±8.584	1.008±0.231
MILLION [9]	6.337±5.990	0.432±0.343	12.18±3.772	0.937±0.240	14.31±7.907	1.167±0.529	9.621±5.814	0.622±0.404	5.134±3.191	0.344±0.195	15.20±13.39	0.865±0.739
DF (proposed)	10.67 † ±7.650	0.542 ‡ ±0.391	20.59 † ±0.485	0.998 ‡ ±0.038	18.05 † ±6.940	1.285‡ ±0.485	13.57 † ±6.301	1.092 ‡ ±0.373	9.982 † ±9.851	0.693 † ±0.682	16.58 † ±16.78	0.528 ±0.541

4.1 Experimental Setup

Datasets and Evaluation Metrics. To ensure a comprehensive assessment, we utilize six real-world market datasets containing daily open, high, low, and close price data. In some cases, we extend the time span or increase the number of stocks using publicly available resources; in others, we obtain data from online financial providers [43]. Each dataset is chronologically split into training, validation, and test periods in a 7:1:2 ratio, with specific date ranges and asset counts summarized in Table 2.

We evaluate performance using standard financial metrics [12]. Annualized Rate of Return (ARR) measures profitability, while Annualized Sharpe Ratio (ASR) evaluates risk-adjusted return by dividing ARR by return volatility, with higher values indicating better performance. To evaluate risk control, we report Maximum Drawdown (MDD) — the largest peak-to-trough loss — and Annualized Volatility (Avol) — the standard deviation of returns on a yearly scale — both lower is better. Additionally, we include Calmar Ratio (CR) [51], which relates return to drawdown, and Sortino Ratio (SoR) [40], which adjusts return by downside deviation. Higher CR and SoR reflect superior profit-to-risk trade-offs.

Table 2: Summary of six real-world market datasets.

Dataset (# Assets)	Training Period	Validation Period	Testing Period
U.S. stock market ¹ (224)	1993.08-2014.01	2014.02-2016.12	2017.01-2022.11
KR stock market ² (528)	2002.01-2016.07	2016.08-2018.08	2018.09-2022.11
CN stock market ³ (34)	2010.01-2017.09	2017.10-2018.10	2018.11-2020.12
JP stock market ⁴ (118)	2014.09-2021.02	2021.03-2022.01	2022.02-2023.12
U.K. stock market ⁴ (21)	2015.01-2021.03	2021.04-2022.02	2022.03-2024.01
Crypto market ⁵ (37)	2018.10-2023.01	2023.02-2023.11	2023.12-2024.10

¹ <https://wrds-www.wharton.upenn.edu>² <https://finance.yahoo.com>³ <https://github.com/TradeMaster-NTU/TradeMaster>⁴ <https://datalab.snu.ac.kr/dtml/#datasets>⁵ <https://binance.com>

Baselines. We compare our method against classical portfolio optimization approaches, recent two-step generative models, and modern deterministic deep learning and reinforcement learning methods. Classical baselines include Autoregressive Integrated Moving Average (ARIMA) [5], Cross-Sectional Momentum (CSM) [36], and Buying Winner and Selling Loser (BWSL) [19]. We also include a BuyHold strategy, which mimics the overall market behavior by assigning equal weights to all assets in the portfolio. Contemporary deterministic methods include MetaTrader [34], which leverages meta-policy for dynamic strategy selection, EarnMore [55], which adapts the investment universe over time, and MILLION [9], which continuously adjusts risk using a pivot portfolio derived from a

min-variance model. All of these methods generate a single deterministic portfolio per input. In contrast, recent two-step generative approaches include FactorVAE [11], which applies a Variational Autoencoder (VAE) [24] in a factor-model context, D-Va [26], which incorporates a diffusion process within a hierarchical VAE [44], and TMDM [29], another diffusion-based model.

Model Architecture. We use the Adam optimizer [25] and determine the initial learning rate for each dataset via grid search over {5e-3, 1e-3, 5e-4}. We set the maximum training epochs to 400 and the batch size to 128, applying early stopping based on the validation loss defined in Eq. (18). The auxiliary loss weight λ is set to 1.0, and the look-up window L is 256. For the diffusion process, we select the timestep T by searching over {100, 300, 500}. In experiments involving stochastic models, we generate 50 samples per test instance, using the average performance of these samples as the final performance measure.

4.2 Quantitative Assessment (RQ1)

Table 1 presents an extensive evaluation of DIFFOLIO against various baseline methods, covering classical, deep generative, and state-of-the-art deterministic deep learning (DL) and reinforcement learning (RL) approaches. We assess performance across six real-world market datasets using ARR and ASR, reporting mean and standard deviation ($\mu \pm \sigma$) over 50 generated samples for stochastic models. Deterministic models either omit variance or present an average over 10 independent runs with different random seeds.

The results clearly demonstrate that DIFFOLIO substantially outperforms classical strategies by considerable margins in both profitability and risk-adjusted performance across most market conditions. These findings underscore the advantage of directly modeling portfolio distributions rather than relying on manually crafted rules.

Compared to existing two-step generative methods, DIFFOLIO achieves statistically significant improvements in almost all datasets, highlighting the effectiveness of end-to-end training and direct risk conditioning. While profit-focused deterministic neural baselines often surpass DIFFOLIO in profitability criterion, this is expected given that DIFFOLIO is specifically designed for balancing risk and return. Its reported ensemble performance across multiple risk levels inherently reflects a more balanced strategy. The relative underperformance of DIFFOLIO on the crypto dataset stems from the extreme volatility of cryptocurrency markets. Because reported results average across five risk levels, they obscure the strength of low-risk variants, which are better suited for such conditions.

Table 3: Ablation study of the proposed method, DIFFOLIO (DF), and its variants at different risk levels γ_i , evaluated using ARR and ASR metrics. The best results are in bold, and the second-best are underlined.

Dataset	U.S. stock market		KR stock market		CN stock market		JP stock market		U.K. stock market		Crypto market	
Methodology	ARR(%) $\uparrow$	ASR $\uparrow$	ARR(%) $\uparrow$	ASR $\uparrow$	ARR(%) $\uparrow$	ASR $\uparrow$	ARR(%) $\uparrow$	ASR $\uparrow$	ARR(%) $\uparrow$	ASR $\uparrow$	ARR(%) $\uparrow$	ASR $\uparrow$
DF- γ_0	13.26 $\pm$ 6.236	0.674 $\pm$ 0.319	20.31 $\pm$ 0.114	1.006 $\pm$ 0.005	15.57 $\pm$ 5.475	1.358 $\pm$ 0.485	10.77 $\pm$ 1.379	1.088 $\pm$ 0.119	8.687 $\pm$ 8.961	0.716 $\pm$ 0.739	29.77 $\pm$ 14.42	0.958 $\pm$ 0.481
DF- γ_1	13.77 $\pm$ 7.613	0.701 $\pm$ 0.391	20.78 $\pm$ 0.183	1.022 $\pm$ 0.008	16.10 $\pm$ 5.789	1.305 $\pm$ 0.474	10.50 $\pm$ 0.415	1.022 $\pm$ 0.037	7.692 $\pm$ 9.238	0.598 $\pm$ 0.719	20.70 $\pm$ 17.09	0.659 $\pm$ 0.550
DF- γ_2	10.55 $\pm$ 6.425	0.535 $\pm$ 0.328	<u>21.14</u> $\pm$ 0.026	1.030 $\pm$ 0.013	18.00 $\pm$ 6.372	1.373 $\pm$ 0.487	<u>14.19</u> $\pm$ 0.192	1.191 $\pm$ 0.017	7.252 $\pm$ 7.686	0.546 $\pm$ 0.579	11.74 $\pm$ 14.68	0.369 $\pm$ 0.462
DF- γ_3	11.09 $\pm$ 5.806	0.562 $\pm$ 0.296	20.91 $\pm$ 0.003	1.005 $\pm$ 0.009	<u>20.16</u> $\pm$ 6.312	<u>1.383</u> $\pm$ 0.447	13.08 $\pm$ 1.035	0.931 $\pm$ 0.073	12.55 $\pm$ 10.46	<u>0.828</u> $\pm$ 0.696	12.65 $\pm$ 8.287	0.392 $\pm$ 0.260
DF-nRG (γ_4)	4.673 $\pm$ 8.267	0.236 $\pm$ 0.419	19.81 $\pm$ 0.006	0.925 $\pm$ 0.028	20.43 $\pm$ 8.763	1.006 $\pm$ 0.425	19.32 $\pm$ 12.02	<u>1.228</u> $\pm$ 0.786	13.73 $\pm$ 10.77	0.777 $\pm$ 0.622	8.016 $\pm$ 17.88	0.261 $\pm$ 0.575
DF- γ^*	12.03 $\pm$ 6.935	0.623 $\pm$ 0.355	21.72 $\pm$ 0.025	1.063 $\pm$ 0.011	18.62 $\pm$ 5.993	1.486 $\pm$ 0.480	14.16 $\pm$ 0.268	1.229 $\pm$ 0.025	<u>13.32</u> $\pm$ 9.224	0.889 $\pm$ 0.712	31.31 $\pm$ 14.83	0.990 $\pm$ 0.477

Table 4: Risk comparison of DIFFOLIO (DF) and baselines. Best results are in bold. Symbols $\dagger$ and $\ddagger$ denote statistically significant improvements ($p < 0.05$, two-tailed t-test) over the best two-step generative and deterministic baselines, respectively.

Dataset	U.S. stock market				KR stock market				CN stock market				JP stock market				U.K. stock market				Crypto market			
Methodology	MDD	AVol	CR	SoR	MDD	AVol	CR	SoR	MDD	AVol	CR	SoR	MDD	AVol	CR	SoR	MDD	AVol	CR	SoR	MDD	AVol	CR	SoR
BuyHold	34.6	0.17	0.17	0.61	34.7	0.23	-0.04	-0.26	18.1	0.17	0.36	0.64	10.4	0.15	0.83	0.81	9.16	0.12	0.52	0.89	25.0	0.32	1.14	1.27
ARIMA [5]	35.3	0.09	-0.03	-0.19	17.5	0.08	-0.08	-0.25	54.4	0.22	-0.63	-2.47	7.02	0.11	0.99	1.22	15.7	0.22	-0.12	-0.25	55.6	0.33	-0.21	-0.64
BWSL [19]	28.9	0.16	0.06	0.17	24.8	0.19	0.34	1.57	29.8	0.17	-0.31	-0.77	10.1	0.11	1.34	2.47	31.8	0.21	-0.30	-0.75	42.8	0.42	-0.06	-0.08
CSM [36]	29.0	0.10	0.23	0.63	50.0	0.23	-0.20	-0.62	14.1	0.11	2.59	2.94	8.53	0.15	1.77	1.65	13.6	0.17	-0.11	-0.15	59.5	0.35	-0.24	-0.73
FactorVAE [11]	40.3	0.13	-0.09	-0.48	21.1	0.12	0.27	0.71	44.2	0.19	-0.57	-2.06	27.7	0.12	-0.50	-1.94	18.6	0.15	0.36	0.62	18.1	0.03	1.14	1.57
D-Va [26]	19.6	0.06	-0.05	-0.25	10.5	0.05	0.03	0.11	9.12	0.08	1.39	2.37	4.41	0.04	1.11	0.78	9.12	0.08	0.19	0.22	14.6	0.19	1.23	1.46
TMDM [29]	12.6	0.06	0.05	0.16	17.0	0.07	0.15	0.56	17.3	0.13	0.38	0.69	6.12	0.06	0.74	1.10	17.9	0.12	0.06	0.05	31.1	0.22	0.19	0.48
MetaTrader [34]	51.2	0.38	0.23	0.62	38.0	0.17	0.22	0.86	45.8	0.34	0.42	1.35	37.7	0.30	0.46	1.43	17.4	0.20	0.56	0.94	46.1	0.31	0.71	1.89
EarnMore [55]	42.9	0.30	0.32	0.83	36.3	0.23	0.51	0.94	30.9	0.29	0.73	1.59	29.9	0.29	0.50	0.75	14.0	0.16	0.80	1.24	37.5	0.39	1.05	2.13
MILLION [9]	36.6	0.15	0.17	0.59	29.8	0.13	0.41	0.74	36.6	0.12	0.17	0.59	9.81	0.17	0.98	0.86	8.92	0.15	0.58	0.86	31.7	0.18	0.48	1.23
DF (proposed)	32.0 $\ddagger$	0.20	0.42 $\ddagger$	1.01 $\ddagger$	24.0 $\ddagger$	0.21	0.86 $\ddagger$	1.70 $\ddagger$	10.4 $\ddagger$	0.14	2.04 $\ddagger$	2.62 $\ddagger$	8.67 $\ddagger$	0.12 $\ddagger$	1.71 $\ddagger$	2.38 $\ddagger$	13.9	0.14 $\ddagger$	0.91 $\ddagger$	1.32 $\ddagger$	24.6 $\ddagger$	0.32	0.87	1.03

As shown in Table 3, low-risk configurations (e.g., DF- γ_0) achieve higher performance than high-risk ones while remaining competitive with deterministic baselines.

Table 4 further illustrates DIFFOLIO’s robust capability in risk management. It attains superior performance in risk-adjusted metrics such as CR and SoR, indicative of strong profit-risk trade-offs, alongside competitive outcomes in MDD and AVol. The results confirm DIFFOLIO as a robust risk-aware portfolio optimization framework that achieves strong returns under user-defined risk.

4.3 Ablation Studies (RQ2)

To further analyze the effectiveness of risk conditioning, Table 3 presents an ablation study of DIFFOLIO (DF) under different settings. DF- γ_i represents performance at each predefined risk level γ_i , while DF-nRG refers to a variant that excludes the risk guidance mechanism introduced in Section 3.3, effectively setting $\zeta^{(v)} = 0$. Another variant, DF- γ^* , considers a *foresight* scenario where risk levels γ_τ^* are dynamically selected at each trading step τ , assuming an expert trader with perfect knowledge of market risk conditions.

To obtain γ_τ^* , we evaluate the quantile of the future absolute index return r_τ' using L past absolute index returns. A high quantile indicates high future volatility, in which case a lower risk level is selected for γ_τ^* as the optimal decision. Specifically, the quantile q_τ of r_τ' at each step τ is computed as:

$$q_\tau = F_{R_\tau'}(r_\tau') = \frac{1}{L} \sum_{l=1}^L \mathbf{1}(R_{\tau,l}' \leq r_\tau'), \quad 0 \leq q_\tau \leq 1, \quad (22)$$

where F is the empirical CDF, $R_\tau' \in \mathbb{R}^L$ denotes L historical absolute index returns, and $\mathbf{1}(u \leq v)$ is an indicator function. The quantile q_τ is then discretized into $\gamma_{\max}$ risk levels as follows:

$$\gamma_\tau^* = \min(\lfloor \gamma_{\max} (1 - q_\tau) \rfloor, \gamma_{\max} - 1), \quad (23)$$

where lower γ_τ^* values correspond to lower risk.

The results in Table 3 confirm that conditioning on different risk levels is both effective and essential. For instance, DF- γ^* achieves a higher ASR, highlighting that dynamically adjusting risk levels in response to market conditions can be beneficial for risk management.

Table 5: Summary of rank score r_s for DIFFOLIO, evaluated with and without risk guidance (DF-nRG).

Methodology	U.S.	KR	CN	JP	U.K.	Crypto
DF-nRG	0.459	0.649	0.544	0.488	0.464	0.379
DF (proposed)	0.681	0.949	0.854	0.943	0.867	0.617
Improvement (%)	48.37	46.22	56.99	93.24	86.85	62.80

However, the advantage of dynamic selection varies depending on market characteristics. In the highly volatile crypto market, selecting a low risk model results in superior performance, whereas in the stable U.K. stock market, a higher risk model leveraging fewer assets performs better. These results emphasize that training models at multiple risk levels and adapting them based on market conditions is crucial for effective risk-aware portfolio optimization.

4.4 Effectiveness of Risk Guidance (RQ3)

To evaluate how the proposed risk guidance (Section 3.3) influences the denoising process, we measure changes in the *rank scores* r_s of portfolio risk both with and without the guidance mechanism. Specifically, we use a normalized Spearman’s rank correlation [42] between the ranks of observed portfolio risks $R[\rho'(\mathbf{w}^{(v)})]$ and the target risks $R[\gamma]$, where $R[\cdot]$ denotes the ascending rank function, i.e., $R[x] = k$ if x is the k -th smallest element in the set. Thus, $r_s = 1$ denotes perfect alignment of observed and target risk orderings, while $r_s = 0$ denotes a completely reversed ranking.

Table 5 presents the computed r_s values for DIFFOLIO (DF) and its variant without risk guidance (DF-nRG) across multiple datasets. The results indicate that incorporating risk guidance during denoising consistently enhances alignment between generated portfolio risk and target risk levels. By explicitly guiding the denoising process, DIFFOLIO more effectively refines portfolio risk, producing allocations that adhere more closely to the desired levels.

4.5 Visualization of Denoising Paths (RQ4)

To enhance interpretability, we visualize how DIFFOLIO’s denoising diffusion process evolves to achieve the intended risk measure at different targeted risk levels. Briefly recalling the generation

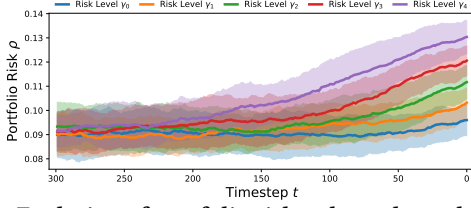


Figure 3: Evolution of portfolio risk ρ throughout the denoising process for different target risk levels γ_i in the U.K. stock market dataset. The solid lines represent the mean portfolio risk across 50 independent sampling trajectories, while the shaded regions indicate standard deviations.

process: we begin with a fully noisy portfolio $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \sigma_x^2 \mathbf{I})$ and perform T reverse steps until arriving at the final denoised portfolio $\hat{\mathbf{x}}_0 \approx \mathbf{x}_0$. At each reverse step t , we define the reverse transition for $\mathbf{x}_{t-1}$ by predicting a denoised portfolio $\hat{\mathbf{x}}_\theta$, which is used to estimate the reverse transition mean $\mu_\theta(t)$ in Eq. (20).

Now, let $\mathbf{x}_t^{(\tau)}$ denote the predicted portfolio at trading step $\tau \in [0, T_d]$ and reverse step $t \in [0, T]$. Then, for each chosen reverse step t , we collect all predicted portfolios $\mathbf{x}_t^{(\tau)}$ over the entire trading horizon, compute their returns, and evaluate the risk measure. Repeating this at every risk level γ leads to $\gamma_{\max} \times T$ measurements of portfolio risk in total. To account for stochastic variability, we generate 50 independent sampling trajectories; for each step t and risk level γ_i , we plot the mean of the resulting risk values as a bold line and shade the standard deviation in the same color around it.

Collectively, Figure 3 illustrates how the risk measure ρ evolves throughout the denoising process for each target risk level γ with $T = 300$, demonstrated using the testing period from the U.K. stock market. Early on, the portfolios start off fully random, so there is little distinction across risk levels. However, as t decreases, the portfolios begin to differentiate themselves, more precisely converging toward the intended risk level and occupying a lower risk region if so targeted. This finding confirms that adjusting the mean of the sampling distribution by incorporating the user-specified risk level indeed yields a more fine-grained trajectory for denoising.

4.6 Risk-Driven Weight Allocation (RQ5)

To further demonstrate DIFFOLIO’s ability to adapt its allocations according to different target risk levels, we examine how the model assigns higher weights to high-volatility assets as the risk level increases. Let $S^{(\tau, \gamma)}$ be the set of asset indices corresponding to the top- k largest absolute returns at trading step τ according to Eq. (16). For each asset $j \in S^{(\tau, \gamma)}$, we record its absolute weights at all risk levels, denoted by $\mathbf{w}_j^{(\tau)} = \{|\mathbf{w}_j^{(\tau, 0)}|, \dots, |\mathbf{w}_j^{(\tau, \gamma_{\max}-1)}|\}$. We then measure the normalized Spearman’s rank correlation $\tilde{r}_s(\mathbf{w}_j^{(\tau)})$ between the ranks of weight magnitudes $R[\mathbf{w}_j^{(\tau, \gamma)}]$ and the target risks $R[\gamma]$. A value of $\tilde{r}_s(\mathbf{w}_j^{(\tau)})$ close to 1 denotes strong risk compliance. To evaluate this aggregately, we define the *risk conformity*:

$$\bar{r} = \frac{1}{T_d k} \sum_{\tau} \sum_{j \in S^{(\tau, \gamma)}} \tilde{r}_s(\mathbf{w}_j^{(\tau)}). \quad (24)$$

Table 6 reports the risk conformity $\bar{r}$ for $k = 10$ across multiple datasets. In most cases, the model shows high conformity, implying strong alignment between weight assignments and user-defined risk levels. Markets such as JP and CN, which exhibit lower return

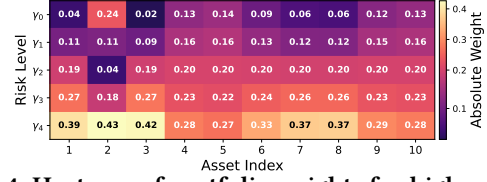


Figure 4: Heatmap of portfolio weights for high-volatility assets in the U.S. stock market on 2018-04-19. Allocations increase with risk level, demonstrating adaptability of DIFFOLIO to user-defined risk preferences.

Table 6: Risk conformity $\bar{r}$ of DIFFOLIO across the markets.

U.S.	KR	CN	JP	U.K.	Crypto
0.514	0.570	0.502	0.497	0.556	0.549

volatility, yield moderate conformity. Figure 4 illustrates this behavior; each column of the heatmap shows the column-wise normalized weights $\tilde{\mathbf{w}}_j^{(\tau)}$ for a single asset $j \in S^{(\tau, \gamma)}$ in the U.S. stock market on 2018-04-19. We observe that, as the target risk level increases, the allocation to that asset consistently rises. Overall, these results confirm that DIFFOLIO effectively adapts its weight distributions in response to different risk preferences.

5 Related Work

Deterministic Models for Portfolio Optimization. Classical approaches to portfolio optimization involve factor models [6, 13] or the Markowitz model [31], yet they struggle to capture complex, nonlinear asset interactions [22, 37, 49]. Deep learning approaches improve predictive modeling capabilities for portfolio optimization [20, 35, 41, 47, 50, 53], yet deterministic methods remain limited by their inability to generate diverse portfolios, restricting adaptability across varied market conditions and risk preferences.

Probabilistic Models for Portfolio Optimization. To address this limitation, deep generative models enable sampling of multiple portfolios [8, 11, 29]. Most works follow a two-stage process — forecasting future time series before applying predefined rules, such as Markowitz optimization [26, 30], equal weighting [21, 27], or top- k asset selection [8, 38]. However, since these models are not directly optimized for portfolio performance, their generated allocations may not fully align with portfolio optimization objectives.

6 Conclusion

In this paper, we propose DIFFOLIO, a risk-aware portfolio optimization framework based on diffusion model. DIFFOLIO learns a pseudo-optimal portfolio distribution for direct generation, leveraging correlated asset embeddings. We introduce a risk-aware denoising network for risk-conditioned generation and a risk guidance mechanism for fine-grained risk control. Experiments across multiple datasets show that DIFFOLIO achieves the state-of-the-art performance, with strong alignment between generated portfolio risks and user-specified targets. Future works will explore latent diffusion models to further enrich the solution space.

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